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Integration of Nanoengineering with Artificial Intelligence and Machine Learning in Surface-Enhanced Raman Spectroscopy (SERS) for the Development of Advanced Biosensing Platforms

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Surface-enhanced Raman spectroscopy (SERS) has emerged as a powerful tool for biomedical diagnosis, combining heightened sensitivity with molecular precision. The integration of artificial intelligence (AI) and machine learning (ML) has further elevated its capabilities, refining data interpretation, pattern prediction, and bolstering diagnostic accuracy. This review chronicles advancements in SERS diagnostics, emphasizing the collaboration between ML and innovative nanostructures, substrates, and nanoprobes for SERS enhancement. The breakthroughs are highlighted in SERS-based point-of-care techniques and the nuanced detection of key biomarkers, from nucleic acids to proteins and metabolites. The article also addresses prevailing challenges, such as the need for standardized SERS methodologies and optimized platforms. Moreover, the potential of portable SERS systems is discussed for clinical deployment, as well as current efforts and challenges in clinical trials. In essence, this review positions the fusion of nanoengineering, AI, ML, and SERS as the frontier for next-generation biomedical diagnostics.

1. Introduction

The detection accuracy of biomolecules or chemicals with high sensitivity is crucial in clinical diagnosis. This is a salient aspect of diagnostics regarding diseases such as cancer and Alzheimer's, where early detection can improve chances of recovery and survival. However, biomarkers that confirm the onset of the disease are often present at very low concentrations, reflecting the disease progression stage. In biomedicine, conventional methods for detecting these biomarkers typically involve sophisticated laboratory-based instruments, microscopy, cell culture, proteomics, or molecular biology-based techniques that require skilled personnel, dedicated laboratory space, and significant time.[1,2] Although these methods are very sensitive, they can take days to detect an analyte and are generally unsuitable for use in remote or emergency settings. Therefore, there is a need for instruments and detection methods that are rapid, sensitive, and portable for easy transportation and operation.[3] Since the first electrode for biosensors was developed by Leland C. Clark in 1953, the technology has made significant progress and continues to do so today. This progress can be attributed to advancements in both the biological and physical components of biosensors.[4] Biosensors are analytical tools that combine biological components, such as antibodies, with a physicochemical or optical detector to achieve high selectivity. Their primary purposes are to detect, identify, and quantitatively determine molecules to monitor biological processes and diagnose diseases.[5] In recent years, biosensors have gained significant attention as a field of research due to their rapidity, affordability, high sensitivity, selectivity, and ease of use. These advantages have led to advances in personalized medicine and the point-of-care (POC) detection of disease markers. One of the most sensitive and potent spectroscopic techniques for biosensors is SERS, which offers a million-fold improvement in the Raman signals when employing the proper SERS substrates compared to traditional Raman spectroscopy.

1.1. Surface-Enhanced Raman Spectroscopy-Based Biosensing

Raman spectroscopy is a molecular vibration-based method that can be used to identify chemicals and their molecular structures, permitting an exceptionally subtle analytical sensing system. Despite not being initially regarded as a sensitive technique, it was confirmed that the Raman scattering signals of analytes can be amplified by employing functionalized materials and engineered probes in nanoscale patterns. As a result, scientists have made significant progress in comprehending the fundamental

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principles of SERS and developing it for numerous applications.[6] The SERS phenomenon, first reported in 1974 by Fleischmann while studying pyridine adsorbed on a roughened silver electrode,[7] has gained significant attention for its potential applications in label-free and multiplex biochemical and biological detections. Since the 1990s, SERS has become one of the most attractive analytical tools for biosensing.[8] Like conventional Raman spectroscopy, SERS is an emission method that relies on inelastic scattering of incident laser light. This process generates spectrum peaks that are frequency-shifted from the incident energy resulting from the molecule's vibrational modes.

SERS biosensors have distinct advantages over conventional biological detection methods, primarily due to their narrow peak width, high sensitivity, and broad sensing range. Therefore, SERS sensors are highly versatile and can be used in various detection scenarios, including food safety, pharmaceuticals, biomolecular detection, viral identification, environmental monitoring, and biomedicine testing.[6,9–11] Recently, SERS has become a crucial research technique in the realm of single-molecule science. As SERS is adopted more extensively, there is great potential for *in vivo* imaging, rapid disease diagnosis, and detection of trace substances. SERS is widely recognized as a technique that can provide detailed structural information of the target molecule through the amplified Raman signal on engineered nanostructures, often referred to as a "fingerprint" spectroscopy. SERS is also considered a multiplex detection technology due to its high spectral resolution, which enables the detection of multiple targets simultaneously.[12]

In biomedical applications, SERS-based biosensing and bioimaging encompass two detection techniques: label-free detection and indirect detection using SERS labels. Label-free detection provides intrinsic fingerprint information of biomolecules with high scattering areas in biomedical samples, whereas indirect detection employs Raman active molecules connected to nanoprobes for highly sensitive SERS detection and imaging. Due to the fact that the majority of biomolecules *in vivo* exhibit a low Raman cross-sectional area, scientists typically employ potent and distinctive Raman molecules to enhance the sensitivity of detection.[12,13]

Despite significant progress in the enhancement mechanism and substrate development, the SERS signal still exhibits low repeatability and susceptibility to substrate effects during analysis. This is a limiting factor in its application to qualitative or semi-quantitative analysis. To expand the applicability of SERS technology, there is a need to develop and improve quantitative analysis methods with higher accuracy. In this regard, researchers have achieved remarkable success on both theoretical and experimental levels by investigating the enhancement mechanism of SERS and developing innovative substrates. Some researchers have achieved ultra-sensitive detection of molecules with single-molecule detection capabilities.[14,15] The continued exploration of the enhancement mechanism has led to the development of numerous novel SERS substrates and probes which have further optimized SERS structures and enabled ultra-sensitive single-molecule response detection. Therefore, designing a unique and target-oriented SERS platform plays a crucial role in qualitative and quantitative analysis thereby increasing the efficacy of the platform's biosensing performance.

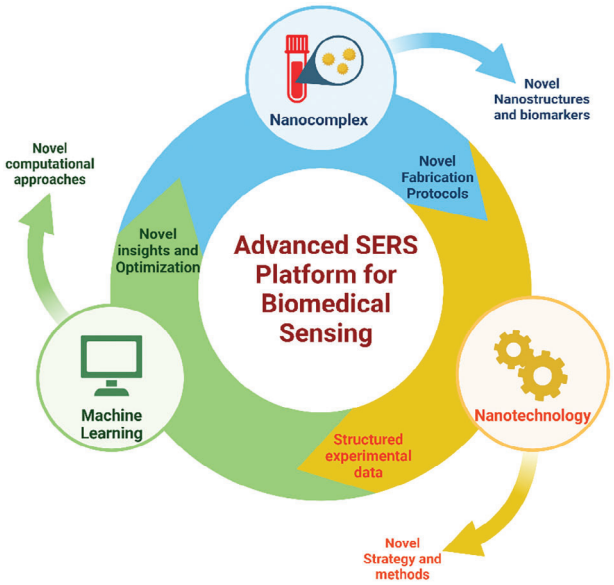

Figure 1. Schematic representation of advanced SERS platforms for biomedical sensing via the integration of nanotechnology, machine learning, and nanocomplex fabrication (created with BioRender.com).

1.2. Advancing Surface-Enhanced Raman Spectroscopy with Artificial Intelligence and Machine Learning

AI and ML have been increasingly applied in SERS to improve the accuracy and efficiency of analysis. ML algorithms such as Support Vector Machine (SVM), Neural Networks (NN), Principal Component Analysis (PCA), and Partial Least Squares (PLS) can handle large and complex data sets and extract features from raw data to make accurate predictions and classifications.[16] These algorithms have been applied to analyze SERS spectra, including PCA, SVM, Convolutional Neural Network (CNN), Artificial Neural Network (ANN), and Random Forest (RF).[16]

In the field of medicine, ML combined with SERS has been used for disease classification, diagnosis of infectious diseases and cancer, detection of metabolites, and the creation of medically relevant databases.[16] ML algorithms can analyze SERS spectra to distinguish between different diseases, bacteria, and biomarkers. They can also identify different stages of the same disease and different types of the same disease.[16] ML algorithms have also been used to enhance the SERS signal. For example, PCA has been used to reduce the dimensionality of the data and extract useful information, significantly enhancing the SERS signal.[17]

AI and ML have been applied to surface-enhanced spectroscopies, such as SERS and surface-enhanced infrared absorption spectroscopy (SEIRA), to overcome technical limitations and improve their performance.[18] These techniques have been widely used in various fields, including biology, medicine, electrochemistry, catalysis, and materials science.[18] The combination of ML, AI, and SERS has the potential to revolutionize various fields by providing rapid, accurate, and non-invasive analysis (**Figure 1**).

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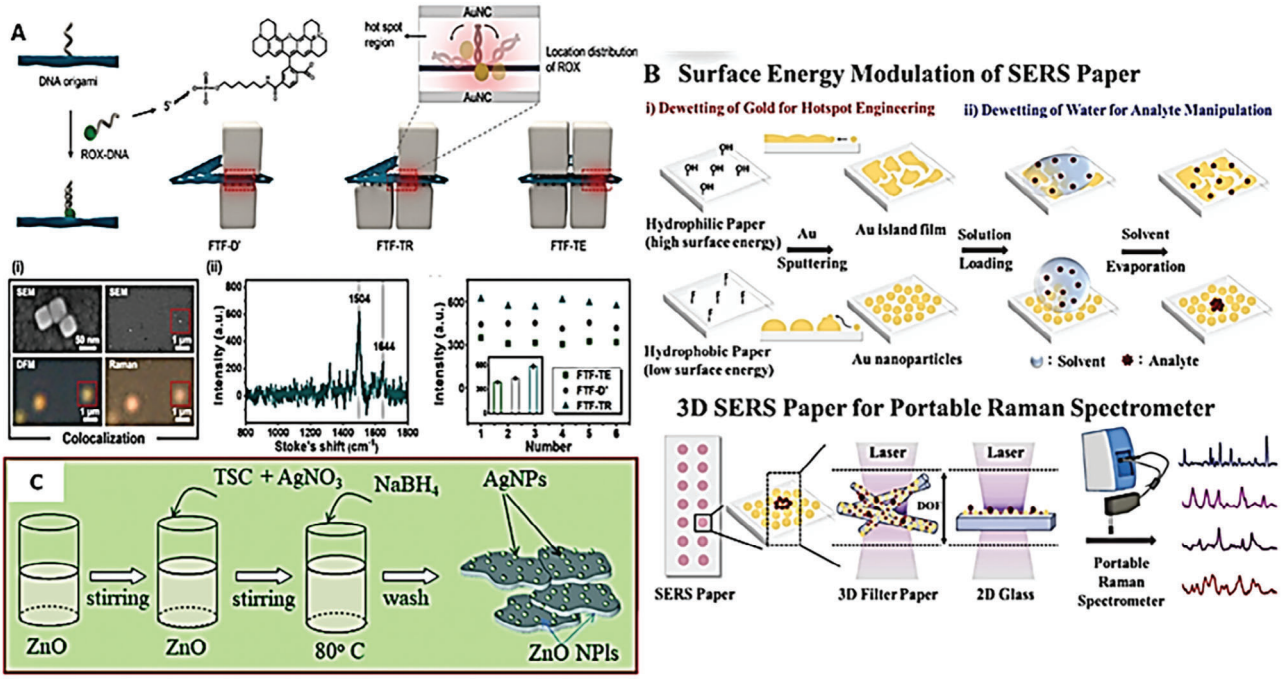

Figure 2. Schematic of different enhancement strategies for SERS. A) Hotspot engineering, schematic illustration of AuNC nanostructures with three Raman probe ROX molecules at different sites. ROX-DNA hybridized with the DNA origami template in the AuNC nanostructure's hot spot region. Reproduced with permission[15] copyright 2022 American Chemical Society. B) Analyte manipulation combined with Hotspots. Fabrication of 3D SERS paper. FOS modification of the filter paper enables the direct formation of sputtered AuNPs for hotspot engineering, along with a hydrophobic condensation effect for analyte manipulation, dramatically enhancing the Raman signals. Reproduced with permission[20] 2023 copyright Elsevier. C) Hybrid SERS platform. Illustration of ZnO/Ag hybrid preparation process. Reproduced under the terms of the CC BY-NC[21] 2022 Royal Society of Chemistry.

2. Design of Nanoengineered SERS-Based Biosensors

It is important to consider the engineering of nanomaterials and nanostructures for highly sensitive SERS biosensors.[19] These substrates can be categorized into different types based on their design, enhancement protocols, type of molecules, and fabrication strategies. Below, we provide an overview of the various SERS design strategies and substrate fabrication methods that are useful for different biomedicine applications (**Figure 2**).

2.1. Strategies for SERS Signal Enhancement

SERS is a phenomenon that significantly amplifies the Raman signals of analytes near the surface of materials. This amplification is mainly attributed to two types of enhancement: electromagnetic and chemical. Electromagnetic enhancement results from the amplification of Raman scattering by electromagnetic fields on the surface of the enhancing material: localized surface plasmon resonance (LSPR). This is due to the excitation of surface-electron oscillations having an enhancement factor (EF) of 10^6 to 10^{12} on the SERS platform.[12,22,23] Chemical enhancement occurs through charge transfer between the analyte molecule and enhancing substrate, which changes the molecule's polarizability and enhances its Raman scattering. It was verified that most semiconductor-enhanced Raman scattering involves a charge transfer process between the substrate and target molecules, presenting a novel approach to exploring the chemical mechanism of SERS. The chemical mechanism effect, which is responsible for the interfacial charge transfer interaction, confers exceptional applications on semiconductor SERS substrates, enabling the observation of chemical processes, predicting photo-voltaic efficiency, and investigating surface science. These benefits highlight the crucial role of semiconductor SERS substrates indicating the application of non-metallic materials as exceptional SERS substrates.[12,24] The combination of both types of enhancement provides SERS with significantly improved sensitivity reaching single-molecule detection while maintaining the main benefits of Raman spectroscopy.[12]

The EF and the analytical enhancement factor (AEF) are two commonly used metrics to measure the overall SERS enhancement resulting from electromagnetic and chemical effects. The EF is expressed through Equation 1 and is used to measure the signal enhancement by comparing the intensity (I) of SERS and normal Raman scattering (N) signals.[25] This parameter serves as a valuable benchmark for evaluating SERS and determining the average field enhancement experienced by each molecule across various platforms and is crucial to accurately determine the number of molecules (N) measured within the laser excitation volume for this metric.

$$EF = \left(I_{SERS} \times N_N\right) / \left(I_N \times N_{SERS}\right) \quad (1)$$

$$AEF = \left(I_{SERS} \times C_N\right) / \left(I_N \times C_{SERS}\right) \quad (2)$$

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The AEF is expressed through Equation 2 and takes an analytical approach to measuring signal enhancement by relating signal intensity to the concentration of the analyte (C) instead of the number of molecules. Its application is especially valuable in situations where estimating the number of analyte molecules present is challenging, particularly for analytes that do not have a specific affinity to plasmonic surfaces.[26] Computational simulations have the capability to calculate the enhancement factor instead of relying on algebraic methods. In a computational study, researchers employed the finite element method in COMSOL Multiphysics to investigate the electromagnetic field (EM) enhancement of gold nanoparticles (AuNPs) with varying shapes.[27] This investigation included single and dimer forms of the NPs with simulation parameters using a standard-size mesh at an incident wavelength of 785 nm. They found that the resonance condition between the incident wavelength and spherical AuNPs resulted in the highest EM field enhancement compared to other NP shapes although the overall EM field enhancement factor is low at 10^2. However, in the case of dimer combinations, pyramid-shaped NPs demonstrated the maximum enhancement factor of 10^6 with a 1 nm inter-particle gap due to the coupling effect of bonding dipole plasmon at hotspot regions. As the inter-particle distance increases, the EM enhancement subsequently decreases.[27] This type of study illustrates the crucial aspect of the design of SERS substrates for achieving maximum enhancement factors using inter-particle gaps when the morphology of noble metal nanoparticles is insufficient to bridge the gap. This, however, is just one of many examples of the aspect of design in enhancement factor maximization.

The most common strategy in developing SERS platforms is hotspot engineering: a set of new techniques that produce unique plasmonic structures in 0D and 1D hierarchies; and two and three-dimensional plasmonic metacrystals. These structures have remarkable plasmonic properties and greater field enhancement with their applications in various strategies developed to improve SERS design through analyte manipulation. These strategies focus on the fundamental working principles and fabrication techniques and how they enhance SERS detection.[26,28] One report indicates that combining SERS platforms with new functional materials, such as 2D materials and semiconductors, can create hybrid ensembles with controllable electromagnetic and chemical properties augmenting performance[12] (Figure 2). In the next section, we discussed these strategies briefly.

2.1.1. Hotspot Engineering

Raman signal enhancement can be achieved by placing molecules in what are known as "hotspots" located between nanoparticles, where the electromagnetic field is maximized as a result of the interaction between the LSPR of neighboring nanoparticles. In SERS research, the application of 0D to 3D materials in the nanoscale focuses on electromagnetic hotspot engineering to achieve intense light confinement. 0D/1D approaches create nanoparticles with unique structural features to enhance hotspot densities at the single particle level. In contrast, multidimensional plasmonic platforms exploit plasmonic coupling along the x-, y-, and/or z-axis to significantly increase hotspot intensities and enable more efficient utilization of SERS platforms within 3D laser excitation volumes.

Not only are researchers focusing on creating noble metal nanostructures with unique shapes, but there is also a growing interest in developing three-dimensional SERS substrates featuring periodic structures. For example, researchers created highly reliable and sensitive structures using ordered micro-pyramid arrays and silver nanoparticles, producing an EF of 8.8×10^8 and facilitating the precise detection of the interest molecule.[29] The use of structures like these is gaining widespread attention for their increased surface area and higher concentration of hotspots. Another effective approach is the combination of plasmon elements and metal cavities. In a recent study, Au hexagonal cone arrays (Au-HCAs), fabricated using uniform porous anodized aluminum oxide templates and prepared an Ag film with a cone cavity through thermal deposition, were used for trace detection through SERS. The Au-HCAs were then prepared through electrodeposition growth technology with an optimal electrodeposition time of 600 s yielding the highest SERS activity. The Au-HCAs had a limit of detection of 10^{-9} M for R6G, with high reproducibility and uniformity at 10^{-6} M, demonstrating good stability. Additionally, the Raman intensity had a good linear correlation with molecular concentration, indicating that the Au-HCAs had good quantitative analysis ability.[30] Moreover, a study by Russo et al. showed that hollow AuAg nanoshells (AuAg NSs) were utilized to improve the Raman signal of mercaptobenzoic acid (4-MBA), allowing for readout via SERS. It was observed that the AuAg NSs provided greater Raman enhancement compared to Au nanostars.[10]

When the gap between NPs is below 10 nm, the interstices of nanostructures or hotspots can experience a significant increase in the EF of SERS, ranging from 10 to 10^4, due to the production of new gap plasmon resonances. To achieve this, the controlled aggregation of plasmonic colloids, with analytes located in hotspots, is critical for producing ultra-sensitive SERS sensors that work in liquid environments.[28] For example, by adjusting the concentration of Au precursor, the nanogaps between AuNPs on the SiO_2 surface in SiO_2/Au core-shell nanostructures were controllable, ranging from 4.16 to 0.98 nm, leading to the development of efficient SERS hotspots. The resulting SiO_2/Au core-shell nanostructures with a 0.98 nm gap exhibited a significant SERS enhancement factor of $\approx 3.8 \times 10^6$ when stimulated by 785 nm photoexcitation. These nanoprobes showed detectable *in vivo* SERS signals at a concentration of 16 μg mL≈ in animal tissues at a depth of 7 mm.[31] In the interstices between the nanoparticles, the presence of water helps prevent the colloidal system from collapsing, which would cause a significant reduction in the EF due to quantum plasmonic effects. Research has shown that using "wet" plasmonic aggregates can be more effective than dried ones, allowing for attomolar sensitivity to be reached.[28]

Hotspot engineering for SERS presents several challenges that necessitate attention and resolution.

Fabrication techniques pose a primary challenge in the creation of well-defined and reproducible hotspots with nanometer-sized gaps. Methods such as electron beam lithography, focused ion beam milling, and nanosphere lithography have been employed, offering a high level of control and reproducibility,[32] but they encounter limitations in terms of scalability and cost.

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Achieving precise control over gap size and geometry at the nanoscale is challenging, resulting in variations in SERS enhancement factors.

The stability and lifetime of hotspots are significant concerns as well. Hotspots are typically formed by bringing plasmonic nanoparticles in close proximity, but maintaining their stability over time can be difficult due to nanoparticle aggregation, oxidation, or degradation. Furthermore, hotspots may have limited lifetimes due to factors such as thermal effects and surface diffusion.

Reproducibility is another critical challenge in hotspot engineering. Nanoscale fabrication techniques often yield inconsistencies in gap sizes, particle positions, and local environments, leading to variations in SERS signals between different hotspots. Ensuring high reproducibility is essential for practical SERS applications. Compatibility with analyte molecules is a crucial consideration in hotspot engineering. Some molecules may encounter hindrance or electrostatic interactions that limit their access to the hotspots, affecting overall SERS performance. Hotspot engineering should strive to be compatible with a wide range of analyte molecules to enable broad applicability.[26]

The potential solution pathways are emerging to address these challenges. AI/ML-driven approaches have shown significant promise. Deep learning algorithms can now predict optimal substrate geometries and fabrication parameters by analyzing correlations between structural features.[33] Real-time ML algorithms provide feedback during measurements, allowing dynamic adjustment of processing parameters to maintain consistency that could be able to balance enhancement and reproducibility.[34] Also, template-assisted fabrication methods combined with automated quality control systems offer another powerful solution. These approaches use precisely engineered templates, such as anodized aluminum oxide or lithographically patterned surfaces, to guide the assembly of plasmonic nanostructures into highly ordered arrays with controlled gap sizes and orientations.[32,35] Moreover, integrating real-time characterization tools, such as scanning electron microscopy or atomic force microscopy, directly into the SERS measurement setup can provide detailed feedback on the substrate morphology and hotspot distribution. This allows for better monitoring and control of the substrate quality, potentially improving reproducibility. This systematic approach to substrate fabrication and characterization provides a robust foundation for reliable SERS measurements in practical applications. However, achieving optimal EFs necessitates a deep understanding of the intricate electromagnetic interactions involved, which heavily depend on specific design parameters.

2.1.2. Analyte Molecules Manipulation

Improving the SERS detection of analytes can be achieved through efficient analyte manipulation is vital for analytes with no specific affinity to the plasmonic surface, especially for those with low Raman cross-sections. In this approach, researchers address the challenges associated with poor analyte affinity to the plasmonic surface proposing innovative approaches to confine analytes near electromagnetic hotspots and SERS-active regions for optimal SERS sensitivity and signal reproducibility.[28,36] The latest analyte manipulation techniques involve the following approaches: utilizing chemical interactions to capture and/or direct analyte molecules to functional plasmonic surfaces, physically confining analytes near the SERS-active region, and, encapsulating plasmonic nanostructures in a secondary sorbent material to concentrate analytes directly over the electromagnetic hotspots.[26]

Furthermore, modifying analyte molecules chemically can also enhance their intrinsic Raman cross-sections thereby increasing their corresponding SERS signals. This can be achieved through multifunctional hybrid SERS platforms which combine engineered plasmonic nanostructures with secondary functional materials. The integration of these materials allows for a synergistic modulation of both the chemical and electromagnetic properties of the resulting hybrid SERS platforms and/or analytes, enabling efficient SERS responses. There are new and developing ideas for manipulating analyte molecules that can significantly improve SERS performance by 10 to 10^4 times.[26] The sample manipulation can be incorporated with hotspot engineering to obtain a synergetic effect on the EF of SERS. In a recent study, combination strategies were employed for creating 3D fibrous substrates to design a SERS platform.[20] This approach allows for the direct formation of a monolayer of AuNPs on filter paper at a large scale. The process starts by coating the fiber surface with perfluorooctyltrichlorosilane (FOS) to manipulate the surface energy. During subsequent sputter deposition, the significant difference in surface energy between the Au atoms and the FOS-treated fibers promotes the formation of discrete AuNPs, leading to the creation of numerous hotspots. Additionally, the paper decorated with AuNPs maintains its hydrophobic properties, allowing the analyte molecules manipulation to concentrate near the hotspots after the solvent evaporates. Moreover, the 3D fibrous structure offers a spacious detection area, effectively utilizing the wide depth of field of a portable Raman spectrometer. These characteristics make the designed SERS paper highly sensitive, with an EF of 1.63×10^8 and a LOD in the nanomolar range when employing a portable 785 nm Raman spectrometer.[20]

Despite the advantages, this approach also faces challenges. One challenge is the complexity of modifying analyte molecules without affecting their inherent properties or altering the analyte-target interactions, especially for complex biological samples. Careful consideration is required to ensure that the modifications do not compromise the specificity and sensitivity of the biosensor. Additionally, the manipulation of analyte molecules can be technically demanding and time-consuming, requiring specialized techniques such as surface functionalization, self-assembly, or external manipulation methods. These techniques may introduce additional steps and complexities to the assay workflow, potentially hindering the practical implementation of the approach.

2.1.3. Hybrid SERS Platforms

To improve the sensitivity of SERS, additional materials such as graphene, semiconductors, and piezoelectric polymers have been added to common metallic nanoparticle-based SERS platforms. Combining these materials can provide chemical and electromagnetic enhancement and improve the detection limits by 10 to 10^2 times compared to conventional metallic NP-based SERS platforms.[37] Hybrid SERS platforms thus

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integrate the properties of both materials to overcome the limitations of traditional noble metal-only SERS platforms, such as enhanced analyte affinity near plasmonic surfaces, increased stability, and higher reproducibility.[26] Instead of relying on LSPR, non-noble materials typically enhance signals through structural effects and charge transfer interactions with analytes.[38] For example, Zinc oxide (ZnO) nanostructures as a semiconductor material in hybrid SERS substrates, like nanotubes, nanorods, and nanowires, create intense electric fields at specific geometric features, such as their tips or edges, which act as hotspots for signal enhancement.[12,39] It offers advantages such as high refractive index, high chemical stability, low cost, and biocompatibility. The incorporation of ZnO into SERS substrates can enhance the plasmonic properties and improve the overall surface enhancement of the Raman signal.[12] In this regard, Adesoye et al. developed colloidal semiconductor-based substrates for SERS application in biomedicine by Mg-doped ZnO. They demonstrated that the maximum SERS enhancement was obtained at a 2% doping concentration without any significant cytotoxicity.[40] TiO_2 nanostructures, such as nanospheres, nanorods, and nanotubes, can confine light within their unique geometries, creating high electric field regions that enhance Raman scattering. TiO_2 also offers additional signal enhancement through charge transfer interactions with analytes, facilitated by its bandgap and surface states.[41,42] Jiao et al. presented a highly efficient near-infrared (NIR)-SERS biosensor based on a hybrid semiconductor, which includes bimetallic Au/Ag NPs decorated on well-aligned TiO_2 nanorod arrays (NRAs).[43] The hybrid sensor exhibited a stronger absorption capacity in the 400- nm region, resulting in significantly higher NIR-SERS activity compared to TiO_2 NRAs decorated with monometallic Au or Ag NPs. The developed NIR-SERS sensor has the potential for ultrasensitive detection of antibiotics like ciprofloxacin and chloramphenicol in real-world water samples with detection limits as low as 10^{-9} M and 10^{-8} M, respectively, using 785 nm laser excitation.[43] Similarly, Wang et al. enhanced the signal of a SERS platform by coating a thin layer of graphene oxide on the $Ag/SiO_2/Au/Si$ substrate. They achieved a SERS performance with 1.1×10^5 analytical EF and a LOD of 10^{-9} M for melamine detection.[44] Fabrication techniques for hybrid SERS platforms include depositing ultrathin Au films onto different types of substrates, including semiconductor materials, graphene, etc.[45] Other techniques include hydrothermal synthesis, atomic layer deposition, and pulsed laser deposition, which allow for the control of film thickness, surface roughness, and overall plasmonic enhancing capabilities.[46]

Hybrid SERS platforms offer numerous advantages in terms of sensitivity and reliability. However, they come with a set of challenges that need to be carefully addressed. These challenges include material compatibility, complex fabrication processes, signal uniformity, and stability. Integrating different materials in hybrid SERS platforms requires meticulous material selection and optimization to ensure compatibility and avoid undesired interactions or degradation. The fabrication of hybrid SERS platforms involves precise control over material deposition, alignment, and patterning, which can be a complex process.[46] For instance, the maximum SERS EF was achieved for Au films near the percolation threshold for all types of substrates. The thickness of the percolation threshold was found to be between 5 and 7 nm for different substrates.[45]

Techniques like lithography, self-assembly, or chemical synthesis introduce their own challenges that need to be overcome for reliable and reproducible results. Uniform signal enhancement throughout the SERS substrate is crucial for accurate measurements, but variations in material properties or fabrication processes can cause discrepancies in signal intensity or spatial distribution. For example, the use of graphene in hybrid structures can suppress the luminescence background by an average of 40%, which can affect the SERS signal.[45] Achieving consistent enhancement and uniformity poses a significant challenge that requires attention. Stability and longevity are essential for practical applications of hybrid SERS platforms. Factors like material degradation, component loss, or changes in material interfaces can affect long-term performance. Therefore, careful material selection, protective coatings, and optimization of fabrication processes are necessary for ensuring platform stability and reliability.

Addressing these challenges requires collaboration among scientists and engineers from various disciplines, such as materials science, chemistry, nanotechnology, and device engineering. By overcoming these challenges, hybrid SERS platforms hold immense potential for applications in chemical sensing, biomedical diagnostics, environmental monitoring, and more.

2.2. Design of SERS Nanosubstrates and Nanoprobes

From a practical standpoint, an ideal SERS platform should not only concentrate on SERS enhancement strategies, but also on combining high sensitivity, high stability, recyclability, and cost-effectiveness. Among all of the mentioned parameters, stability is a major challenge faced by SERS nanoprobes when operating in biological systems or environments. Biological systems exhibit dynamic conditions such as temperature fluctuations, pH variations, and the presence of numerous proteins, which can impact the stability of the nanoprobes.[47,48] Temperature changes can cause structural changes or aggregation of the nanoprobes, affecting their SERS performance. pH variations may alter the surface chemistry of the nanoprobes, leading to changes in their signal enhancement properties. The presence of proteins can result in interactions or adsorption on the nanoprobes' surfaces, potentially hindering their stability and affecting their SERS activity.[49] Overcoming these stability challenges requires careful design and engineering of SERS nanoprobes to ensure robust performance in complex biological systems. For efficient biomarker detection, SERS nanotags need to accurately identify targets and provide specific signals for readout. The design of SERS nanotags typically involves four essential components: a plasmonic nanostructure, a Raman reporter, a protective shell, and a targeting unit.[50]

The Raman enhancement based on LSPR exhibits a distance-dependent feature due to the electromagnetic field dampening effect as a function of distance increase with effective signal enhancement typically limited to ≈10 nm near the surfaces of the nanostructures. Plasmonic nanostructures are usually composed of noble metals, such as Au, silver, and copper, that possess strong LSPR properties. Moreover, plasmonic nanostructures with different morphologies exhibit diverse LSPR-related Raman signal enhancements as they have varying electromagnetic field

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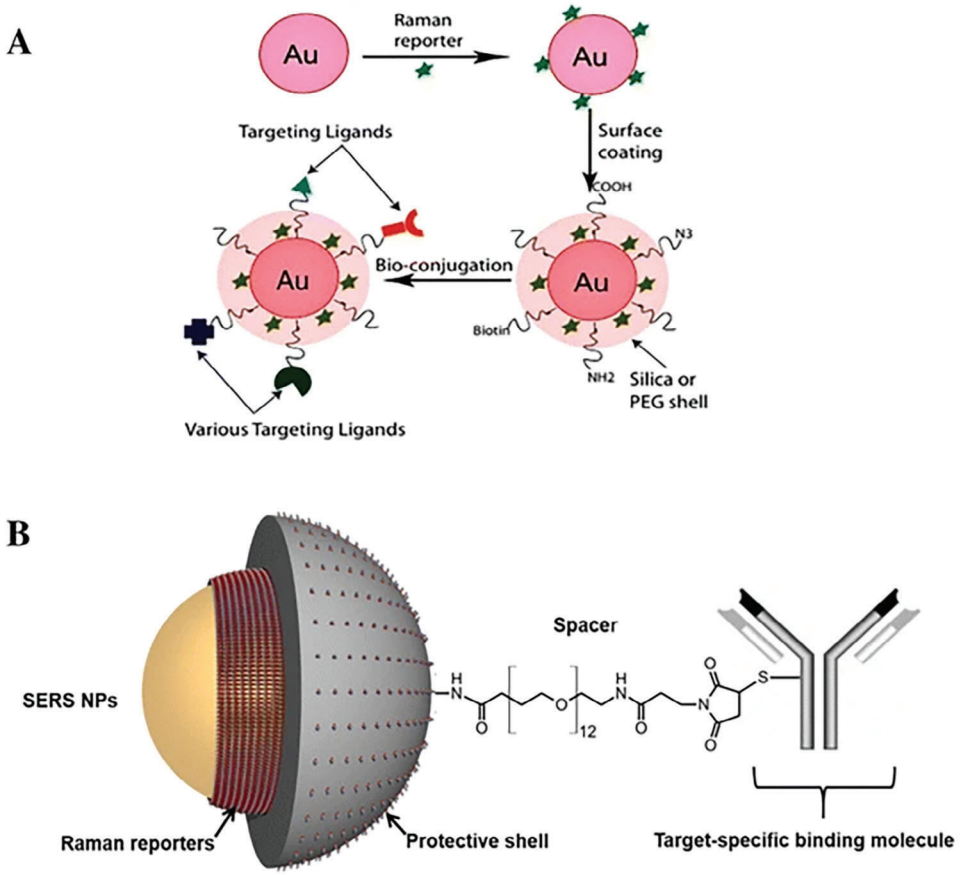

Figure 3. Typical SERS nanotags, consisting of a metal nanoparticle core, Raman reporter molecules (dye as Raman reporter in A and self-assembly monolayer of the small molecule as Raman reporter in B) on the metal surface, a biocompatible layer, and targeting ligands. Cited from[52] copyright 2018 the Springer Nature. Reproduced with permission[32,54] copyrights 2015 American Chemical Society and 2009 WILEY-VCH GmbH.

distributions. For example, isotropic plasmonic nanostructures, such as nanospheres, perform differently than anisotropic structures, such as nanostars and nanoflowers, in enhancing Raman signals.[23,51] Based on these various morphologies, promising Raman probes for *in vitro* and *in vivo* imaging can be designed. SERS nanoparticles, also known as SERS nanotags, are commonly used for biomedical applications.[50,52] These particles typically consist of a plasmonic nanoparticle core coated with molecules known as Raman reporters, which are encapsulated in a protective shell that can be decorated with targeting ligands (**Figure 3**).

The Raman reporters are chosen not only for their polarizability and resonance properties but also for their binding affinity to the metal NP surface, which increases the overall number of reporters per particle and generates a higher signal intensity. The metal/molecule system is then encapsulated in a biocompatible coating that stabilizes the metal nanoparticle and prevents desorption of the Raman reporter molecules from the NP's surface. Finally, the resulting SERS nanotags can be functionalized with targeting ligands that direct the nanoparticles toward specific disease markers, enabling their detection using SERS-based biosensing and imaging.[53]

The Raman reporter provides intrinsic molecular fingerprint information, which can serve as a unique characteristic signal to identify targets. Due to the narrow width of Raman peaks, different Raman reporters with non-overlapping characteristic bands can label multiple targets simultaneously. Raman reporters are divided into three categories: dye-based, thiol-based, and triple bond-based molecules. For highly sensitive detection of individual biomarkers, dye-based Raman reporters, such as crystal violet, are recommended over simultaneous analysis of multiple biomarkers. Thiol-based molecules (e.g., 4MBA) are favored as Raman reporters in multi-biomarker analysis due to their easy functionalization on nanostructures via sulfur and Au/silver interactions with few characteristic peaks that minimize Raman peak overlaps. Triple bond-modulated molecules are gaining attention as the next generation of Raman reporters for detecting multiple biomarkers. Altogether, these molecules have unique and straightforward Raman signals beyond 1800 cm^{-1}, situated in the Raman silent region, and are less likely to interfere with biological samples.[50,55]

The choice of Raman reporter is just as important as the plasmonic core in designing an effective nanotag. The selection of an appropriate Raman reporter depends on the surface properties of the metal nanoparticle core, including atomic composition (e.g., Au, Ag, Pt), zeta potential (positive or negative), and the density of the stabilizing agent. In general, desirable properties of any Raman reporter include a high Raman scattering cross-section

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(usually found in conjugated electron-rich polyenes or aromatic molecules like dyes), high photo- and chemical stability, molecular symmetry (to minimize the number of peaks in a Raman spectrum while increasing their relative intensity), absorption maximum in resonance with the excitation laser wavelength, and the presence of anchoring groups, such as thiols, disulfides, thiophenes, isothiocyanates, etc., with high affinity to the nanoparticle surface.[53]

Another important factor in fabricating SERS nanoprobe is the coating agent. To ensure the stability of the metal-molecule system and prevent desorption of Raman reporters from the metal core, an encapsulant layer as coating is typically used. The encapsulant layer acts as a protective shield against environmental factors that could decrease colloidal stability. The most commonly used encapsulants include silica,[56] synthetic polymers,[57] and biomolecules such as proteins or lipids.[58,59] Depending on the application, a composite shell could be used, which can include a combination of these components.

2.3. Detectable Biomarkers via SERS Biosensors

The main goal of medical diagnosis has always been to detect diseases earlier to increase the chances of the successful applications of treatment. Biomarkers are measurable indicators of normal or pathological biological processes or medical treatment responses and are crucial in early disease detection. Biomarkers can be invasive, detected through tissue biopsies, or non-invasive, detected through liquid biopsies.[60] Due to their high sensitivity and specificity, SERS biosensors have become an attractive platform for detecting various biomolecules, including proteins, nucleic acids, and small molecules. They have been widely used for the detection of proteins due to their ability to provide label-free and highly sensitive detection. For example, Lu et al. developed a SERS-based immunoassay for the detection of prostate-specific antigen (PSA) and thrombin simultaneously, achieving a LOD of 0.01 ng mL^{-1} and 0.01 nM, respectively by using a nanocone array covered with deposited gold film fabricated on glass slides that could be directly sealed with PDMS microfluidic chip as SERS substrates.[61] Similarly, Phuc et al. reported a SERS-based sandwich immunoassay for the detection of α-fetoprotein (AFP) with a LOD of 3.8×10^{-4} ng mL^{-1} using hybrid core-shell nanostructure of Fe_3O_4 (core) and gold (shell).[62]

SERS biosensors have also been utilized for the detection of nucleic acids, including DNA and RNA. For example, Chen et al. reported a SERS-based biosensor for the detection of miRNA, a potential biomarker for lung cancer, with an LOD of 0.1 fM using a core-satellite nanostructure with gold nanorod array as core and Au@4-MBN@Ag nanoparticles as satellite for the SERS platform.[63] Additionally, Wu et al. developed a SERS-based biosensor for simultaneous and sensitive detection of three miRNAs (miRNA, miRNA, and miRNA), achieving an LOD of 349 aM for miRNA, 374 aM for miRNA, and 311 aM for miRNA, by means of a SERS nanotag (probe DNA-conjugated DNA-engineered fractal gold nanoparticles) and a magnetic capture substrate (capture DNA-conjugated Ag-coated magnetic nanoparticles).[64]

Small molecules have been detected by SERS-based biosensors, including active agents, metabolites, and biohazardous substances. For example, Sun et al. developed a SERS-based biosensor for the detection of melamine and thiram, as common food contaminants, with a LOD of 7.38 μg L^{-1} and 86.1 μg L^{-1}, respectively, using hybrid SERS-active substrate comprising of three-dimensional gold nanodendrites decorated by silver nanoparticles.[65] Similarly, Geng et al. reported a SERS-based immunoassay for the detection of amphetamine with a LOD of 2.28 ng mL^{-1} using a high SERS intensity substrate (Au-XP013@Ag) using the hotspot effect and combined it with antibodies to form SERS immunotags (Au-XP013@Ag-AMP-mAb).[66] SERS biosensors have also been used to detect other biomolecules, including carbohydrates and lipids. For example, Jiang et al. developed a SERS-based immunoassay for the detection of carbohydrate antigen 19- with a LOD of 3.43×10^{-4} IU mL^{-1} using nonmetallic substrates comprising 2D molybdenum disulfide as nanoflower and nanosheet.[67] Moreover, Liu et al. reported a label-free SERS-based biosensor for the detection of cholesterol, vitamin D2, and vitamin D3 based on their distinguishing fingerprints demonstrated as stretching vibration signals by using sulfhydryl-modified gold nanoparticles as the SERS substrate.[68]

In recent years, evidence has been mounting to suggest that SERS is a feasible technique for detecting extracellular vesicles (EVs). A recent study presented a label-free and quantitative miRNA sensing platform for detecting urinary exosomal miRNAs using SERS.[69] The platform utilized a 3D hierarchical plasmonic nano-architecture, generating numerous 3D plasmonic hotspots that induce exceedingly high amplification of SERS signals. The 3D SERS biosensor achieved an impressive sensitivity of 10 aM LOD for the target miRNAs (miRa and miR), which was over 1000-fold more sensitive than previously reported miRNA biosensors. Moreover, this method does not require any labeling or pre-treatment steps. Clinical validation using urinary samples demonstrated that the 3D SERS sensor could accurately differentiate prostate cancer patients from healthy controls based on the differential expression level of urinary exosomal miRNAs.[69] Another study developed a new model of SERS to detect hepatocellular carcinoma (HCC) by identifying circulating tumor-derived exosomes.[70] The proposed SERS substrate consisted of etched nano-Au on a glass substrate with innovative dimensions of 200 × 200 nm. The sensitivity of RS examination for detecting HCC was 95% and the specificity was 100%, which was higher than that of the traditional HCC biomarker AFP.[70] Further, the integration of microfluidic and SERS has been used to detect multiple plasma-derived exosome biomarkers of osteosarcoma, a type of bone cancer.[48] The technique enables a very effective way of purifying exosomes and detecting them in their original location with exceptional sensitivity (detecting as few as 2 particles per μL). The entire process takes only 5 h and requires only 50 μL of plasma for each analysis. SERS tags of AuNPs@MBA@anti-CD63, AuNPs@TFMBA@anti-VIM, and AuNPs@MPY@anti-EpCAM were used as Raman reporter. They studied EVs derived from the plasma of 20 osteosarcoma patients and 20 healthy individuals, measuring levels of different protein biomarkers, namely CD63, VIM, and EpCAM. They found significantly elevated levels of these biomarkers in exosomes from osteosarcoma patients compared to those from healthy individuals.[48]

In labeled and label-free approaches, the metallic structure used is crucial for creating effective SERS substrates, and the

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sensitivity of detection is related to the interaction between substrates and optical properties. SERS-tag-based detection requires molecules with narrow Raman signal peaks that can be modified to characterize exosomes. An interesting study used asymmetric core-shell structured Au@Au@Ag nanoparticles using a layer-by-layer self-assembly method and modified them with different Raman molecules and recognition sequences (polyA-DNA) to detect exosomal miRNAs.[71] Another study developed a method to detect three different types of exosomes from breast cancer cells, prostate cancer cells, and colorectal cancer cells simultaneously using different nanoprobes.[72] The nanoprobes consisted of AuNPs functionalized with aptamers and labeled with different reporter molecules, and a capturing substrate was made by binding magnetic beads with aptamer CD63. The SERS fingerprint of exosome phenotypes could be detected, and the method was tested using plasma samples from clinical individuals diagnosed with cancer.[72]

3. SERS-Based Point-of-Care (POC) Diagnostics

POC technologies are portable sensing solutions that play a significant role in advancing diagnostic capabilties. Traditional diagnostic methods can have complicated sample pre-treatment, limited data sources, time, cost, and poor sensitivity. For example, traditional immunoassays are conducted on sophisticated instruments by trained professionals in hospitals with highly time-consuming procedures from sampling to obtaining results.[73] To address these limitations, POC technologies offer the advantage of on-site rapid detection, excellent sensitivity, portability, and ease of operation.[9,74] For example, the use of POC devices can reduce the average time for administering intravenous tissue plasminogen activator treatment from 180 to 90 min, starting from the suspicion of stroke.[75] The previous example showcases the importance of mitigating damage from prolonged wait times of diagnostics for urgentlyneeded therapy for patients.

As the demand for POC biosensors continues to grow in many fields there is an urgent need for the development of rapid and sensitive diagnostic technologies with smart and portable devices. SERS technology is particularly promising in this regard due to its label-free detection, ultrahigh sensitivity, and distinct molecular fingerprinting. For example, finger-pricked blood can be utilized to detect neurological biomarkers at very low concentrations enabling the on-scene triage of traumatic brain injuries.[76] SERS has also been applied to early detection of acute pesticide poisoning and therapeutic drug monitoring.[77] Despite their unique advantages, SERS biosensors have limited use in future POC testing due to their low throughput and reproducibility.[77] Therefore, enhancing the performance of SERS biosensors has become a crucial area of focus in cutting-edge Raman spectroscopy technologies. To address this issue, various techniques have been developed, including SERS combined with microfluidics and SERS-based immunoassays (**Figure 4**). These approaches have shown promise in achieving high throughput detection and better stability.[9] Moreover, the mentioned strategies of SERS enhancement could be applied to develop and design an accurate and sensitive nanoprobe for SERS detection.

3.1. SERS-Based Microfluidic Devices

Microfluidics has emerged as a cutting-edge platform that enables complex and technically challenging laboratory procedures to be carried out on a microchip. By shrinking large experiments into smaller scales, microchips offer several benefits, such as a significant reduction in sample analysis costs, faster experimental turnaround times, and improved analytical tools.[80] Microfluidic chip technologies, including lab-on-a-chip technologies, enable assays to be miniaturized using small fluid volumes that are typically in the micro or nanoliter range allowing for the integration of complex functionalities that meet the demands of POC testing.[81]

Combining SERS features with microfluidic chips could enhance biosensing in various ways: rapid analysis of clinical blood,[82] simultaneous detection of multiple biomarkers,[8] disease monitoring in the early stage,[83] and hybrid application for various diseases. The integration of microfluidic chip technology and SERS technology results in the creation of a microfluidic SERS chip that allows for highly efficient separation and analysis of blood samples. In this regard, Su et al. designed an integrated microfluidic SERS chip that included a separation-decantation cavity, mixing channels, and a SERS detection chamber with an array element. To enhance the SERS signal, a nano-Au on Ag film SERS substrate was embedded in the detection chamber. This allowed for rapid separation and in situ serum testing from different blood samples within a short time frame of 2 min with serum creatinine as their target analyte. The proposed biosensor effectively identified its characteristic peak at 678 cm^{-1}. It was found that the lowest concentration of serum creatinine (LOD) that could be identified was 4.42×10^{-3} μmol mL^{-1} in water and 8.0×10^{2} μmol mL^{-1} with differences in concentration attributable to the complex composition of serum. When blood samples from patients with chronic renal failure and healthy individuals were tested using the microfluidic SERS chip, the results were consistent with those obtained through the traditional enzymatic method.[82]

The microfluidic SERS chips can detect multiple biomarkers at the same time. Moreover, it could be utilized for biosensing of various diseases. Zheng et al. developed a highly sensitive, practical, and simultaneous detection platform to diagnose multiple breast cancer biomarkers, namely CA153, CA125, and CEA, in real human serum samples using SERS-based microfluidic chips.[8] Immuno-Ag aggregates are labeled with different Raman reporters which serves as SERS substrates for detecting multiple breast cancer biomarkers. The use of SERS active substrates in the microfluidic channels improves the sensitivity of the biosensor by two orders of magnitude. The CA153 LOD in buffer solution was found to be as low as 0.0001 U/mL, thanks to the high SERS enhancement effect. Furthermore, the biosensor can detect multiple breast cancer biomarkers in human serum, with the LOD of CA153, CA125, and CEA reaching 0.01 U mL^{-1}, 0.01 U mL^{-1}, and 1 pg mL^{-1}, respectively, all of which are lower than the clinical diagnosis cutoff values.[8] In an interesting study, a microfluidic chip was created, consisting of two side-by-side channels, which enabled the simultaneous detection of CK-MB and cTnI as acute myocardial infarction (AMI) biomarkers using a SERS-based immunoassay.[74] The chip was designed with an

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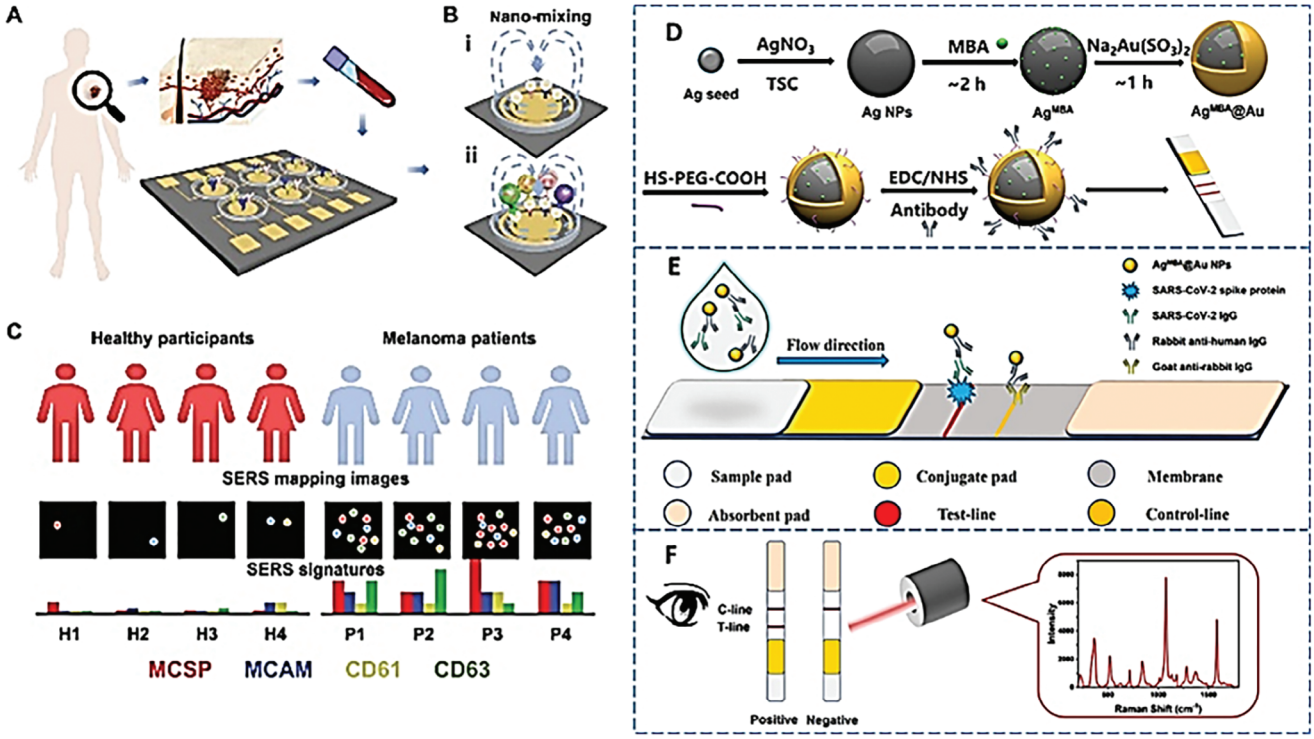

Figure 4. SERS-based POC diagnosis. A-C) Integrated Microfluidic-SERS Platform. A) Serum from a melanoma patient was collected and analyzed by the EPAC-II for EV phenotypic characterization. B) Serum EVs were captured by the anti-MCSP/MCAM/CD61 co-functionalized EPAC-II (i), followed by multiplex SERS nanotag labeling (ii). Both EV capture and labeling processes were performed under nano-mixing. C) On-chip SERS mapping was performed on the clinical cohorts composed of healthy participants and melanoma patients. Reproduced with permission[78] copyright 2021 Wiley-VCH GmbH. D-F) SERS-based immunoassay. D) Flow chart of synthesis and functionalization of Ag^{MBA}@Au NPs. E) Schematic diagram of the test strip structure and detection principle. F) Interpretation of the detection results. Reproduced with permission[79] Copyright 2022 American Chemical Society. The labels on the original figure have been modified to improve readability.

integrated plasmonic nanostripe microcone array that served as the solid capture plate for the immunoassay and a SERS-active substrate. Using the integrated microfluidic chip, they successfully conducted simultaneous trace analysis of CK-MB and cTnI with a LOD of 0.01 ng mL^{-1}[74] Another AMI biosensor was developed by using a single-track finger-pump microfluidic chip for simultaneous SERS detection of cTnI and CK-MB. The designed biosensor demonstrated a high level of sensitivity and portability.[84] The detection process of this microfluidic biosensor was completed in just 5 min without the need for a syringe pump, making it a fast and efficient option. Additionally, the biosensor allows for multiple reagent additions and the removal of unbounded reactants. To enable the simultaneous detection of cTnI and CK-MB biomarkers, they selected two Raman reporter molecules conjugated with AuNPs, malachite green isothiocyanate, and Nile blue, to represent each biomarker. The sensor employs a "sandwich" immunoassay approach that utilizes SERS nanosubstrate, AMI biomarkers, and magnetic beads. The study's findings indicate that the detection limits for cTnI and CK-MB were as low as 0.01 ng/mL, respectively. These LODs are in line with the clinical threshold values for AMI biomarkers.[84]

For monitoring and early-stage diagnosis, SERS-based microfluidic devices can offer high sensitivity. Cao et al. fabricated a pump-free, high-throughput SERS microfluidic chip to detect and trace circulating tumor DNA as a biomarker of gastric cancer.[83] The chip consisted of six parallel functional units, enabling the simultaneous analysis of multiple samples. The chip is assisted by a cascade signal amplification strategy (CHA-HCR), where long-nicked double-stranded DNA can be extended after CHA products trigger HCR. This process results in the accumulation of numerous Cu_2O octahedra on the surface of the highly ordered Au nanobowl array, which generates abundant hotspots to significantly amplify the local electromagnetic field. Using this chip, an ultralow LOD for PIK3CA E542K (1.26 aM) and TP53 (2.04 aM) was achieved, and the whole process was completed within 13 min.[83]

The microfluidic chip can also integrate with SERS-based immunoassay to enhance diagnostic efficiency compared with traditional biosensors. Lu and colleagues developed a new type of solid immunoassay plate and SERS substrate for prostate cancer diagnosis, using a multifunctional nanocone array to perform simultaneous immunoassays of PSA and thrombin using a microfluidic chip.[85] A gold-deposited film was used to create a nanocone array on glass slides, which could be directly sealed with a PDMS microfluidic chip. This substrate was not only SERS active, but also provided a large surface area for the capture aptamer conjugation necessary for forming a "sandwich" immunocomplex. By using this integrated microfluidic chip, it was possible to simultaneously detect PSA and thrombin with detection limits of 0.01 ng mL^{-1} and 0.01 nM, respectively. These results were com-

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pared to other methods such as colorimetric and fluorescent assay that demonstrated this detection strategy had rapid and automatic performance with comparable LOD.[85] The integration of SERS technology with microfluidic platforms has the potential to revolutionize point-of-care diagnostics by offering rapid, sensitive, and multiplexed detection capabilities. These systems not only miniaturize and streamline laboratory processes but also enable the simultaneous analysis of multiple biomarkers, making them ideal for early diagnosis and monitoring of various diseases.

3.2. SERS-Based Immunoassays

SERS has shown great promise in the field of biomedical diagnosis through its application in SERS-based immunoassays. Immunoassays are commonly used in clinical and research settings to detect different biomolecules, such as proteins, antibodies, and antigens.[38,86] In SERS-based immunoassays, a SERS-active substrate is conjugated to an antigen, which binds specifically to its corresponding antibody, generating a SERS signal. SERS-based immunoassays combine the sensitivity of immunoassays with the precise resolution provided by SERS. These assays exploit the specific interaction between antigens and antibodies to enable the identification and quantification of target analytes.[16,49] The SERS effect amplifies the Raman scattering signal emitted by molecules located in proximity to a textured metallic surface, typically comprised of silver or gold NPs. This amplification facilitates highly sensitive detection, often capable of reaching the single-molecule level. SERS-based immunoassays offer notable advantages, including improved sensitivity, the ability to simultaneously analyze multiple analytes (multiplexing), detection without the need for labeling, and compatibility with complex biological samples.[8,87,88] They have found successful applications in diverse domains such as clinical diagnostics, environmental monitoring, and food safety analysis.[88–90] For example, to detect trypsin enzyme, an innovative Raman indicator approach is proposed as a sandwich immunoassay, using a glass capillary-based molecularly imprinted SERS sensor.[91] This sensor integrates a SERS substrate layer for signal enhancement with a polydopamine-imprinted layer for specific protein recognition. Imprinted cavities in the sensor selectively capture target proteins and restrict Raman indicator access to the SERS substrate. Protein binding thus blocks the Raman indicator flow, and the reduction in signal reflects the amount of protein captured. This capillary sensor demonstrated specific and reproducible detection, reaching LOD as low as 4.1 ng L^{-1} for trypsin in biological samples without preprocessing.[91] In another interesting study, a silicon wafer as a substrate for SERS-based sandwich immunoassay has been used to detect human immunoglobulin. The developed sandwich immunoassay showed a LOD of 3 pM.[92] As an example of disease biomarker detection, a highly sensitive and rapid SERS immunoassay was developed for cTnI, a biomarker for diagnosing AMI, using core-shell SERS nanotags. In this approach, a silver shell coated on a gold core bound to the Raman reporter molecule (4MBA) generates numerous "hot spots" with strong Raman enhancement effects, which are then conjugated with antibodies to form SERS immunoprobes.[93] In the presence of cTnI, sandwich structures are formed through immune reactions between the SERS tags and capture probes, with the immune complex enriched via a specific interaction of streptavidin on magnetic beads with biotin. Biomarker concentration is quantified by measuring characteristic Raman peak intensities of the reporter molecules. This platform achieved a detection limit of 9.80 pg mL^{-1} for cTnI, demonstrating high specificity with no cross-reactivity with non-specific proteins.[93]

SERS-based immunoassays offer numerous advantages, but their widespread implementation faces several challenges. Reproducibility and standardization are crucial, requiring protocols and experimental parameters optimization to achieve consistent and comparable results. Signal stability and photobleaching are concerns due to environmental factors and can degrade the reliability of assay outcomes. Multiplexing can be challenging, as spectral overlap and crosstalk between analytes can hinder accurate quantification. Ensuring consistent sensitivity and LOD across analytes and sample matrices is another challenge, influenced by antibody-antigen interactions, background interference, and non-specific binding events. Translating SERS-based immunoassays into real-world applications requires addressing cost, scalability, and integration into existing diagnostic platforms. These challenges necessitate advancements in SERS substrate stability, the development of robust detection strategies, improved assay standardization, and validation in diverse sample matrices.

3.3. SERS-Based Bioimaging

NIR SERS is an emerging bioimaging technique that can provide high-resolution imaging compared to traditional ultraviolet (UV) or visible light excitation, without the potential hazards associated with current imaging techniques such as MRI, CT, and PET. These traditional methods expose individuals to powerful magnetic fields or ionizing radiation over long periods. The success of NIR SERS bioimaging relies on controlled morphology, biocompatibility, and the use of highly sensitive NIR Raman reporters that meet three critical conditions: strong absorption in the NIR region, strong binding to the surfaces of AuNPs, and excellent NIR SERS activity. The creation of controllable nanogaps in AuNPs-assembled nanostructures, which can produce highly enhanced SERS signals at multiple hotspots, is a promising breakthrough.[31,94] In this regard, a new SERS/NIR-II optical nanoprobe named AuDAg2S was developed that combines SERS bioimaging and second region (NIR-II) in-depth biological fluorescence imaging. The nanoprobe consists of Au nanostars, Raman molecular tags, and silver sulfide quantum dots through silica bridges. It has a strong localized surface plasmon resonance and a high photothermal conversion efficiency of 67.1% at 1064 nm, which can effectively kill colon cancer cells. Additionally, the AuDAg2S nanoprobes can enhance the detection sensitivity to the picomolar level and increase the depth of the NIR-II fluorescence signal up to 1 cm under the excitation of the NIR laser.[94] have developed Au-assembled silica nanoparticles as NIR SERS nanoprobes. By controlling the concentration of the Au precursor, they were able to synthesize SiO2@Au@Au nanoparticles with different sizes of AuNPs and nanogaps between them on the silica surface. The nanoprobes with a 0.98 nm gap exhibited a high SERS enhancement factor of $\approx 3.8 \times 10^6$ under 785 nm photoexcitation. These nanoprobes were able to show detectable

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in vivo SERS signals at a concentration of 16 μg mL^{-1} in animal tissue specimens at a depth of 7 mm.[31]

SERS-based bioimaging encounters various challenges that necessitate attention to fully unlock its potential in biomedical applications. One key challenge is achieving adequate sensitivity and specificity to detect and image biomolecules in complex biological environments. The presence of endogenous molecules in biological samples can interfere with the Raman signals of interest. Overcoming this challenge involves developing strategies to enhance the Raman signals of target biomolecules while minimizing interference from background signals. Another challenge lies in the limited penetration depth of Raman signals in biological tissues. The inherently weak Raman scattering process results in signals diminishing rapidly as they penetrate deeper into tissues. This limitation restricts imaging depth and impedes the visualization of structures and events occurring within deeper regions of biological samples. To address this challenge, approaches must be developed to enhance signal penetration depth, such as utilizing specialized probes or advanced imaging modalities. Achieving high spatial resolution is also a significant challenge in SERS-based bioimaging. While Raman spectroscopy offers excellent chemical specificity, its spatial resolution is constrained by the diffraction limit of light.[52,53] Overcoming this challenge necessitates the development of advanced imaging techniques, like super-resolution imaging or near-field techniques, capable of providing sub-diffraction-limited resolution to enable detailed imaging of cellular and subcellular structures. The development of biocompatible and stable SERS probes for bioimaging is another challenge. These probes must be non-toxic, have minimal impact on the biological sample, and exhibit long-term stability to ensure reliable and reproducible imaging results. Furthermore, the probes should be functionalized with targeting moieties to selectively bind to specific biomarkers or cellular structures for targeted imaging. Integration and compatibility with existing imaging platforms and techniques present additional challenges. SERS-based bioimaging should be compatible with common imaging modalities, such as fluorescence or confocal microscopy, enabling multimodal imaging and facilitating the correlation of SERS signals with other imaging data.[95] Seamless integration and compatibility between SERS and other imaging techniques are essential for advancing bioimaging capabilities. Standardization and reproducibility are crucial challenges in SERS-based bioimaging.[52] Variations in imaging results can arise from different SERS substrates, experimental conditions, and imaging protocols. Establishing standardized protocols, benchmarking procedures, and quality control measures is necessary to ensure the reproducibility and reliability of SERS-based bioimaging across various laboratories and research settings. By overcoming these challenges, SERS-based bioimaging holds immense promise for advancing our understanding of complex biological processes, enabling early disease detection, and facilitating personalized medicine approaches.

4. Integrating AI and Machine Learning in Surface-Enhanced Raman Spectroscopy: A Modern Paradigm Shift

AI is increasingly being integrated with SERS to enhance molecular detection across various fields.[96] AI enhances SERS-based biosensors for early disease detection by automating data analysis, recognizing patterns for disease and biomarker identification, and improving sensitivity and specificity through real-time analysis.[97] This integration also supports predictive modeling for concentration estimation and dynamic range expansion and can also handle multivariate analysis to evaluate complex biofluids. Additionally, AI's integration with portable SERS devices provides user-friendly clinical disease detection requiring minimal training while enabling personalized medicine approaches through the development of patient-specific models for monitoring and treatment response evaluation (**Figure 5**).

ML is a subset of AI that focuses on the development of algorithms and statistical models that enable computers to learn from and make predictions or decisions based on data. ML models improve the analysis of SERS spectroscopy by filtering out background noise, thereby enhancing the clarity and accuracy of biosensing data. These models distinguish between relevant signals and noise substantially improving overall data quality. Furthermore, AI facilitates quality control and standardization across different SERS instruments by transferring calibration data from one device to another, ensuring comparable results through learned relationships between measurements. Techniques such as discriminant analysis, ANN, regression analysis, random forest, and k-nearest neighbors are employed to analyze SERS data, providing robust and precise interpretations crucial for reliable biosensing and diagnostic applications.[98]

Discriminant analysis, like Linear Discriminant Analysis (LDA), helps classify data points into distinct categories based on their features. For example, Hu et al. developed a SERS-based sensor for classifying coffee beverages, using discriminant analysis of principal components (DAPC) combined with support vector machine (SVM) and k-Nearest Neighbors (kNN) algorithms resulting in enhanced classification accuracy.[99] This method is also applied in medical diagnostics where Lu et al. developed a multi-receptor SERS sensor to differentiate adenocarcinoma in situ (AIS) from early-stage invasive adenocarcinoma (IAC) using partial least squares discriminant analysis (PLS-DA), achieving 90% sensitivity and 95% specificity.[100,101]

ANNs, inspired by the human brain, consist of layers of interconnected nodes or "neurons" that learn from examples. For instance, Chen et al. used ANN to enhance SERS-based detection with dual SERS aptasensors significantly improving the accuracy of simultaneous detection of multiple targets like chloramphenicol (CAP) and estradiol (E2).[102] Similarly, Zhou et al. employed ANNs to enhance SERS and SEIRA data analysis for molecular diagnostics, improving the identification and quantification of molecular constituents by reducing noise and enhancing spectral resolution.[103]

Regression analysis models the relationship between a dependent variable and one or more independent variables. Garg et al. utilized Gaussian Process Regression (GPR) with SERS to detect SARS-CoV in environmental dust samples, creating a model that accurately quantified virus concentrations and showed strong agreement with actual values.[104] Techniques like Multiple Linear Regression (MLR), Principal Components Regression (PCR), and Partial Least Squares (PLS) reduce data complexity and handle multicollinearity providing robust models for predicting analyte concentrations.

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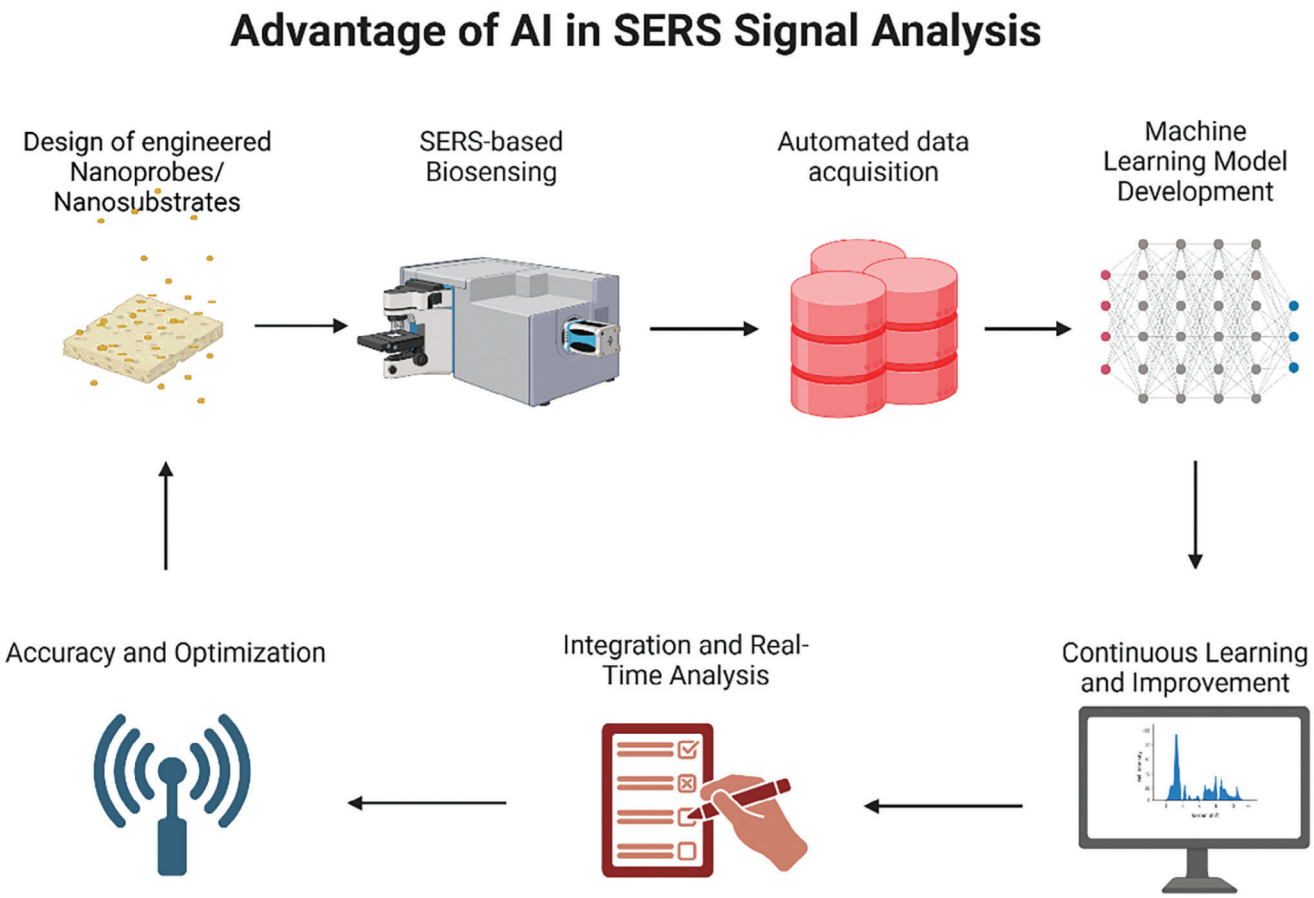

Figure 5. Harnessing the power of AI and ML in SERS signal analysis (created with BioRender.com).

Random Forests involve multiple decision trees working together to make predictions. This method effectively handles noisy data and outliers, making it reliable for classifying and analyzing Raman spectra. Luo et al. aimed to improve the sensitivity and qualitative analysis of polycyclic aromatic hydrocarbons (PAHs) using an ML framework integrated with SERS employing a visual random forest algorithm to achieve 99% accuracy in classifying PAHs.[105]

kNN classifies data points based on the closest examples in the dataset. Hu et al. used kNN algorithms to classify coffee beverages based on SERS spectra, enhancing classification accuracy.[99] This method is also effective for tasks like cancer detection, as demonstrated by Lu et al., who developed a SERS profiling strategy for early lung cancer detection using ANNs and SVM, achieving 83.3% accuracy in classifying plasma exosomes from healthy controls and cancer patients.[106]

In environmental monitoring, Wu et al. developed a high-performance SERS sensing platform using gold nanoparticles on aminated glass capillaries for detecting trace analytes, employing K-means clustering to identify spectral groups, enhancing the accuracy and reliability of SERS analysis. This method was demonstrated to detect Rhodamine 6G (R6G) with high sensitivity.[107]

In contrast to other ML models, CNNs are also used for SERS-based signal analysis. CNNs are designed for handling grid-like data structures and can especially be useful for analyzing spectral images. CNNs were used by researchers after collecting serum samples from patients with different types of cancer and healthy controls and used label-free SERS to obtain Raman spectra from both serum and matched exosome samples.[108–110] Researchers collected serum samples from patients with different types of cancer and healthy controls and used label-free SERS to obtain Raman spectra from both serum and matched EV samples. The obtained spectra were processed using CNN models to classify malignant patients and healthy controls. The study found that the CNN models based on exosome spectra had higher testing accuracies and diagnostic capabilities than the matched serum CNN models.[109] In a retrospective study, they have reported the diagnostic accuracy of using AI to analyze SERS profiles of exosomes to simultaneously detect six types of early-stage cancers (lung, breast, colon, liver, pancreas, and stomach), with promising results in terms of sensitivity and specificity.[111] This suggests that exosomes may be useful as biomarkers for diagnosis and prescreening techniques for psychiatric disorders. Shin et al. developed a new method to diagnose major depressive disorder (MDD) using AI analysis and SERS of plasma exosomes.[112] The system is based on 28000 exosome SERS signals and provides sample-wise prediction results. The method accurately predicted 70 test samples with an area under the curve of 0.939, sensitivity of 91.4%, and specificity of 88.6%. The diagnostic scores were found to be correlated with the severity of depression, showing

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the potential of exosomes as biomarkers for MDD diagnosis and suggesting a new approach for prescreening techniques for psychiatric disorders.[112]

AI and ML significantly enhance SERS-based molecular detection across various fields by improving the clarity, accuracy, and reliability of biosensing data. Techniques like LDA, ANNs, regression analysis, random forest, and kNN provide robust interpretations and facilitate the development of advanced diagnostic applications. For instance, Wang et al. developed an Ag NPs@PDMS sponge SERS substrate for detecting pathogenic microorganisms, achieving high sensitivity and specificity by employing PCA combined with SVC to construct a classification model with 92.17% accuracy.[113]

SERS data processing is challenging due to spectral artifacts, spectral overall, and high volume of spectral data. A hybrid approach using ML models (such as classification, regression, dimensionality reduction, and clustering) and high-volume spectral data are currently used to address these challenges.[103] These models enable automatic design, noise reduction, feature extraction, and dimensionality reduction for time-efficient and reliable data processing/interpretation for disease diagnostics. SERS Spectral data can be processed using the following workflow, generalized for spectral processing using supervised ML models. During data pre-processing raw SERS spectral datasets are streamlined into properly labeled to contain attributes such as sample names and peak intensities.[114] For supervised learning, this spectral data is then separated into the training set and the test set. Data is then processed further to eliminate any missing data, normalizing for instrumentation variation, and evaluating other attributes in the CSV file based on the specific question that the researcher is trying to answer. This data is then processed through ML algorithms such as CNNs. Data is further enhanced by using noise reduction algorithms. A ML model is then developed using a combination of ML algorithms necessary to process and distinguish between various spectral datasets. Finally, evaluation models are deployed, and testing data sets are used to determine the accuracy of the ML model. Data processing and ML phase of the workflow are further manipulated to enhance output accuracy using SERS spectral data.[114,115]

The integration of ML with SERS holds great promise for advancing diagnostics, environmental monitoring, and quality control across multiple sectors, but there are some issues that still need to be addressed. The lack of ground truth and heterogeneity of SERS probes is challenging. However, recent studies have demonstrated that using SERS spectra for diagnosis can provide a wealth of molecular information that can increase diagnostic accuracy.[110] To simplify the detection process and expand its use in clinical practice, it is necessary to develop rigorous standards for sample preparation and portable Raman spectrometers with miniaturization and automation. An ideal platform for future SERS-based detection would integrate purification, recognition, detection, and data analysis.

5. Conclusions and Future Perspectives

SERS has emerged as a promising technique for biomedical diagnosis due to its ability to provide sensitive and specific molecular information. Recent advances in SERS-based biomedical diagnosis systems have shown significant progress in the development of POC devices, strategies for EV-based biomarker detection, and SERS EFs and design. However, challenges remain in the standardization and optimization of SERS substrates, as well as the development of portable and easy-to-use SERS systems for clinical applications.

SERS EFs and design play a crucial role in the sensitivity and specificity of SERS-based biomedical diagnosis systems. Advances in SERS substrates, nanotags, and design strategies have led to improvements in SERS EFs and the ability to detect biomolecules at lower concentrations.

Standardization is a critical aspect of SERS development, particularly for ensuring reproducibility and reliability across different laboratories and applications. The current standardization landscape for Raman spectroscopy, which provides the foundation for SERS standards, is primarily driven by American institutions, notably the American Society of Testing Materials (ASTM International) and the National Institute of Standards and Technology (NIST). ASTM International has established several key standards encompassing basic terminology, calibration protocols, multivariate analysis procedures, and application-specific guidelines. Complementing these efforts, NIST provides certified reference materials that are essential for instrument calibration, performance validation, and quality control measurements.[116]

However, despite these established frameworks, there are significant gaps in standardization specifically for SERS technology that need to be addressed. The rapid pace of SERS technological development has outstripped the evolution of existing standards, leaving crucial areas without standardized protocols. These areas include SERS substrate fabrication and characterization, EF calculations, SERS-specific data processing and analysis methods, and protocols for interlaboratory comparisons. This lack of comprehensive standardization poses a significant challenge to the widespread adoption of SERS technology, particularly in fields requiring high reliability, such as pharmaceutical applications and clinical diagnostics.[116,117]

Moving forward, active engagement from the SERS community is essential to address these standardization gaps. This includes updating existing standards to reflect current SERS capabilities, developing new SERS-specific protocols and guidelines, establishing common metrics for performance evaluation, and creating standardized reporting requirements for SERS experiments. Particular attention should be paid to emerging applications such as portable SERS analyzers and POC diagnostics, where standardization is crucial for ensuring reliable performance across different devices and settings. The development and implementation of comprehensive SERS standards will be fundamental in facilitating the transition of this powerful analytical technique from research laboratories to widespread practical applications.

One of the promising areas of research in SERS-based biomedical diagnosis is the development of POC devices, which enable rapid and accurate diagnosis at the patient's bedside or in resource-limited settings. Recent advances in SERS-based POC technology have focused on the development of low-cost and portable devices that can be used for rapid and accurate diagnosis of various diseases, including cancer, infectious diseases, and neurological disorders. These devices typically use microfluidics and microfabrication techniques to integrate SERS substrates,

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microfluidic channels, and optical detection systems on a single chip. One of the key challenges in developing SERS-based POC devices is the need for reliable and reproducible SERS substrates that can provide consistent and sensitive signals. Additionally, the integration of microfluidic components and optical detection systems requires precise fabrication techniques and optimization of device parameters. In the future, the development of portable and cost-effective SERS-based POC devices is expected to have a significant impact on healthcare.

SERS-based EV-associated biomarker detection has shown great potential to diagnose and monitor various diseases, including multifactorial and infectious diseases. EVs carry specific biomarkers that can be used for early disease diagnosis and personalized treatment. Recent studies have demonstrated the successful detection of EV biomarkers using SERS-based techniques with high sensitivity and specificity. However, challenges remain in developing SERS-based EV detection platforms that are both sensitive and specific to the target biomarkers.

ML and AI are poised to revolutionize SERS biosensing applications. Through the integration of advanced algorithms and data analysis techniques, ML can enhance the sensitivity, specificity, and speed of SERS-based biosensors. AI-driven models can effectively analyze complex spectral data, identify subtle molecular fingerprints, and enable rapid detection and classification of target analytes with high accuracy.

The recent advances in SERS-based biomedical diagnosis systems have shown great potential for improving the accuracy and sensitivity of disease diagnosis and monitoring. Future perspectives include the standardization and optimization of SERS substrates, ML and AI-based biological sample analysis, developing portable and easy-to-use SERS systems for clinical applications, and integrating SERS-based techniques with other diagnostic modalities. Challenges remain in the reproducibility, scalability, and optimization of SERS substrates, as well as the development of SERS-based detection platforms that are both sensitive and specific to the target biomarkers.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

AI, biomarkers, biosensors, diagnosis, ML, nanomaterials, SERS

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